🎯 Quiz: Model Fine-Tuning & LLM Optimization

🧠 Section 1: Conceptual MCQs

1. What is model fine-tuning?

- ⬜ A. Training a model from scratch
- ✅ B. Adjusting a pre-trained model on new data
- ⬜ C. Only changing prompts
- ⬜ D. Increasing model size

2. Which scenario is best suited for prompting instead of fine-tuning?

- ⬜ A. Domain-specific chatbot
- ⬜ B. Repetitive structured output task
- ✅ C. One-time query with general knowledge
- ⬜ D. Training on private dataset

3. What is the main advantage of fine-tuning over prompting?

- ⬜ A. Faster inference always
- ✅ B. Better control over task-specific outputs
- ⬜ C. No need for data
- ⬜ D. Lower memory usage

4. What does LoRA stand for?

- ✅ A. Low Rank Adaptation
- ⬜ B. Linear Representation Adjustment
- ⬜ C. Layer Optimization and Reuse Algorithm
- ⬜ D. Learning Rate Adjustment

5. In LoRA, what is actually trained?

- ⬜ A. Entire model weights
- ⬜ B. Only input tokens
- ✅ C. Small low-rank matrices
- ⬜ D. Output layer only

6. Why is LoRA considered parameter-efficient?

- ⬜ A. It reduces dataset size
- ✅ B. Only input tokensB. It updates only a small subset of parameters

- ⚪ C. It removes layers
- ⚪ D. It compresses input

7. What is the role of a tokenizer in fine-tuning?

- ⚪ A. Train model weights
- 🟡 B. Convert text into tokens
- ⚪ C. Evaluate model
- ⚪ D. Store embeddings

8. Which step comes first in a fine-tuning workflow?

- ⚪ A. Model evaluation
- 🟡 B. Data collection & cleaning
- ⚪ C. Training
- ⚪ D. Deployment

9. What is overfitting in fine-tuning?

- ⚪ A. Model performs well on all data
- 🟡 B. Model memorizes training data but fails on new data
- ⚪ C. Model ignores training data
- ⚪ D. Model becomes faster

10. Which metric is commonly used for classification tasks?

- ⚪ A. BLEU score
- 🟡 B. Accuracy
- ⚪ C. Perplexity
- ⚪ D. Latency

🔍 Section 2: True / False

11. Fine-tuning always requires large datasets.
❌ **False**
(Small datasets can work, especially with LoRA)

12. Prompting does not change model weights.
✅ **True**

13. LoRA modifies all layers of the model.
❌ **False**
(It adds small matrices, doesn't modify everything)

14. Validation data is used to tune hyperparameters.

✅ **True**

15. Fine-tuned models can outperform base models on niche tasks.
✅ **True**

🧩 Section 3: Scenario-Based Questions

16. Scenario:
You need a chatbot specialized in legal documents with consistent tone and formatting.
👉 What should you choose?

- ⚪ A. Prompting
- 🟡 B. Fine-tuning
- ⚪ C. Zero-shot learning
- ⚪ D. Random initialization

17. Scenario:
You want quick results for generating marketing slogans without training.
👉 Best approach?

- ⚪ A. Fine-tuning
- 🟡 B. Prompt engineering
- ⚪ C. LoRA
- ⚪ D. Training from scratch

18. Scenario:
You have limited GPU resources but want to adapt a large model.
👉 Best method?

- ⚪ A. Full fine-tuning
- 🟡 B. LoRA
- ⚪ C. Increase batch size
- ⚪ D. Add more layers

19. Scenario:
Your model performs well on training data but poorly on test data.
👉 What is the issue?

- ⚪ A. Underfitting
- 🟡 B. Overfitting
- ⚪ C. Data leakage

⚪ D. Tokenization error

20. Scenario:
You are comparing two models for summarization.
👉 Which metric is most relevant?

⚪ A. Accuracy
🟡 B. BLEU / ROUGE
⚪ C. Latency only
⚪ D. Token count

⚙️ Section 4: Short Answers

21. Difference between prompting and fine-tuning

- Prompting: You guide the model using input text (no weight change)
- Fine-tuning: You **modify model weights** using training data

22. One benefit of using LoRA
It reduces training cost by updating only **small low-rank matrices instead of the full model**.

23. Two steps in a fine-tuning workflow

- Data collection & preprocessing
- Model training

(Other valid: tokenization, evaluation, deployment)

24. Purpose of validation data
Validation data is used to:

- Tune hyperparameters
- Check model performance during training
- Prevent overfitting

25. One metric used to evaluate LLMs

- Perplexity
  OR
- BLEU / ROUGE
  OR
- Accuracy (for classification)

🧪 Section 5: Practical Thinking

26. Loss not decreasing during fine-tuning — reason?
Possible reason:

- Learning rate too high or too low
- Poor quality data
- Wrong model configuration

27. Why might fine-tuning be expensive?

- Requires GPUs/TPUs
- Large datasets
- Long training time
- High memory usage

28. When should you avoid fine-tuning?

- When task is simple or general
- When you need quick results
- When data is limited
- When prompting works well

29. What happens if tokenizer is mismatched with the model?

- Incorrect token representation
- Poor performance
- Meaning loss in text
- Completely wrong outputs sometimes

30. How does LoRA reduce computational cost?

LoRA:

- Freezes original model weights
- Trains only small additional matrices
- Reduces memory + compute significantly